

Multiscale Adaptive Decoder and Diversity Selection Network for Road Extraction in Remote Sensing Image

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Abstract—Road extraction has been a common and challenging task in the field of remote sensing images. Due to factors such as the high resolution of remote sensing images and the subtle visibility of road features, existing methods often miss certain areas during detection and extraction. These methods struggle to capture contextual information effectively and tend to exhibit false positives (FPs) and false negatives (FNs) when handling objects of varying sizes. This article proposes a network based on a multiscale adaptive decoder and diversity selection network (MADSNet) to address the issue of inadequate contextual information capture. By leveraging feature diverse selection, the method minimizes errors in distinguishing between road features and background interference. Specifically, the multiscale feature flexible extraction (MFFE) decoder utilizes the relevance inquiry attention (RIA) module and scope flexible fusion (SFF) module to enhance the ability to capture contextual information with relatively low computational demands. The optimal choice graph attention (OCGA) module aggregates neighboring nodes with similar features in a graph structure, improving focus on the single class of roads. Furthermore, a multilevel feature selection (MFS) module is proposed to activate the features relevant to the current stage while suppressing features from other stages and interfering with noise. Quantitative and qualitative experimental results on three public datasets demonstrate that the proposed MADSNet outperforms currently popular methods in terms of performance. The code will be available at <https://github.com/Talent02/MADSNet>

Index Terms—Attention mechanism, feature selection, remote sensing images, road extraction.

I. INTRODUCTION

WITH the rapid advancement of remote sensing technology, remote sensing images have become an indispensable source of information in fields such as urban planning, land use, and resource management. These images are typically acquired through satellites, aerial photography, and other means, providing extensive, multispectral, high-resolution geographic data. Deep learning algorithms based on remote sensing images [1] has also become a hotspot, such as remote sensing object detection [2], remote sensing

image classification [3], and remote sensing image change detection [4]. In this remote sensing tasks, road extraction remains a complex and challenging task. Road extraction has broad practical applications, particularly in traffic management and disaster emergency response. Therefore, automated road extraction and updating are crucial. Traditional computer vision methods often struggle to handle variations in road appearances across different scenes and scales. In recent years, the rise of deep learning techniques has brought new hope to the road extraction problem. Deep learning models can learn features and contextual information from images, making them better suited to handle the diversity of road features.

Road extraction methods based on semantic segmentation have gradually become mainstream and have achieved promising results. FCN [5] is one of the early semantic segmentation models. It extends CNN to pixel-level tasks, allowing semantic segmentation of each pixel in the input image, identifying the object or class to which each pixel belongs. Ronneberger et al. [6] first proposed U-Net, an important convolutional neural network architecture widely used in applications such as medical image segmentation, satellite image segmentation, and natural image segmentation. For a high-resolution remote sensing image, some methods [10], [11], [12] are typically able to accurately extract more prominent roads. However, they often struggle with occluded roads or detailed target segmentation, leading to a lack of precision and even missed detections. On the other hand, some methods [13], [14] possess a certain capacity for capturing global contextual information, but are relatively weaker in learning local features.

With the introduction of Transformer [7] into the field of computer vision, their ability to consider information from all positions in a sequence through self-attention mechanism has made them a new technical backbone for road extraction tasks. Many Transformer-based models fail to effectively fuse multiscale features when processing high-resolution images. This limitation hinders the model's ability to fully utilize multiscale information, especially in images with complex structures, thereby negatively impacting segmentation performance. Yang et al. [15] proposed a cross spatial-scale feature interaction module to effectively aggregate full-scale features.

At present, both CNN-based and Transformer-based road extraction methods commonly adopt an encoder-decoder architecture. This structure encodes input information into a latent representation and decodes it by considering both the previous output and the encoded state. It has been widely proven to be effective in segmentation tasks. As the demand

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for higher segmentation accuracy in road extraction increases, models must be capable of perceiving objects at varying scales and eliminating high levels of noise. They also need to analyze fine-grained spatial features and global contextual information. As a result, attention mechanisms have been extensively integrated into deep learning models. For instance, TransRoadNet [16] introduces a context information extraction module (CIEM) to capture the contextual information of roads within an image. The dual attention-selective kernel (DASK) module [17] adaptively focuses on both global context and local salient features.

To address the problem of low segmentation accuracy caused by information loss and incomplete feature extraction, we propose a multiscale adaptive decoder and diversity selection network (MADSNet). This network exhibits a strong capability for capturing contextual information and effectively fuses multiscale features. It enables accurate and robust road extraction from high-resolution satellite remote sensing images. The MFFE decoder in MADSNet offers strong contextual information capture capabilities. This module combines the window mechanism and, in some aspects, mitigates the loss of precision caused by downsampling in the encoder.

Our MADSNet employs a graph attention network (GAT) [18] in its skip connections, which is a specialized form of graph convolution network (GCN) [19]. GCN performs convolution operations on graph-structured data, similar to how traditional convolution neural networks (CNNs) apply convolution operations on regular image data. Previous works [20], [21], [22], [23], [24] have already applied GCN to tasks such as recognition and segmentation of remote sensing images. In graph data, information is represented in the form of nodes and edges, where nodes represent entities and edges represent the relationships or connections between nodes. The convolution operation in GCN takes into account a node's neighboring nodes, enabling nodes to aggregate and propagate information. This helps capture contextual information and relationships between nodes. The goal of GCN is to learn node representations, which can then be applied to tasks like node classification, graph classification, and link prediction. Therefore, we use GCN to predict the classification of each pixel, achieving road segmentation. We employ the k-nearest-neighbors (KNNs) algorithm [25] to determine whether nodes are connected. By selecting the top k nodes with the most similar features to the current node as connected nodes, and treating the remaining nodes as irrelevant, we avoid constructing a graph using all the pixels in the image, thus reducing the computational complexity. The optimal choice graph attention (OCGA) module leverages graph networks to enhance the model's local perception capability, which compensates for the MFFE decoder's limitation of capturing global information.

The multiscale feature selection (MFS) module in MADSNet is specifically designed to address the issue of varying object sizes in remote sensing images. In typical encoder-decoder segmentation models, only the decoded feature map from the final layer is fed into the segmentation head. This approach often results in underutilization of the information from earlier decoding layers, leading to the omission of many fine details during segmentation. In object

detection tasks, feature pyramid network [26] are commonly used to extract multiscale features and construct feature maps at different levels, effectively combining high-level and low-level information to improve detection performance for objects of varying sizes. Similarly, the MFS module in our model consists of four levels, each capable of handling road targets of different sizes. By selectively enhancing or suppressing features on the corresponding feature maps, the module strengthens the segmentation capability at each level. Finally, the segmentation results from all levels are fused together. In summary, the main contributions of this article are as follows.

- 1) We design an MFFE decoder to model the long-term dependencies of features. The RIA module within the MFFE decoder leverages a window mechanism to effectively capture strong contextual information. Additionally, the proposed SFF module expands the receptive field of the MFFE decoder, improving performance while reducing computational overhead.
- 2) We propose an OCGA module that integrates the KNNs algorithm. In the traditional skip connections, we utilize graph convolution to locally aggregate neighboring nodes.
- 3) We introduce the MFS module, which performs feature selection across multiple levels. It activates or suppresses features at individual levels and ultimately enhances the final predictions through multilevel feature fusion.
- 4) A novel deep learning framework (MADSNet) is proposed, which effectively models both multiscale long-range and short-range dependencies. MADSNet demonstrates strong segmentation performance for high remote sensing images (HRSIs) road extraction across three public datasets.

II. RELATED WORK

A. CNNs in Road Extraction of HRSIs

Early traditional methods primarily relied on road geometry, topology, texture, color, and other features, along with some machine-learning-based approaches [27], [28], which were only suitable for handling simple and small-scale road extraction tasks. Today, CNN-based methods dominate the field of semantic segmentation. During classification, FCN [5] did not adequately consider the relationships between pixels, resulting in a lack of spatial consistency and causing fragmented or incomplete object shapes in the segmentation. To address this issue, U-Net series [6], [39], [40] introduced an encoder-decoder symmetric structure, with skip connections being one of its key innovations. These connections link the feature maps of the encoder and decoder to compensate for the loss of spatial information in the high-level features of the decoder, helping the network better localize and segment objects.

However, the fixed-size convolution kernels limit the network's ability to extract features from large and irregular objects. Each version of DeepLab [12], [42], [43], [44] provides some core ideas that address the shortcomings of U-Net. For example, in DeepLabv1, dilated convolutions were introduced to expand the receptive field, improving the network's

ability to extract features from large objects. DeepLabv2 combined the dilated convolutions from v1 with pooling operations to propose the ASPP module, which enables the network to integrate multiscale information. Furthermore, Yi et al. [29] used multiple parallel branches with different receptive fields of dilated convolutions combined with GAP to capture global information.

In recent years, GCN [19] has gradually gained popularity in image tasks [30], [31], [32]. Through the message-passing mechanism, GCN can fuse local and global information from nodes to better capture the dependencies and contextual information between nodes. For example, Zhou et al. [46] used a deep-separable graph convolutional network for segmentation to capture global background road information from both channel and spatial features, improving the expressive power of road features. Furthermore, KSPGAT [47] proposed the SPIN module. This module constructs a spatial graph and an interaction spatial graph, and uses graph inference to extract dependencies and other contextual information between different spatial regions.

B. Transformer in Road Extraction of HRSIs

Similar to GCN, Transformer [7] is a revolutionary neural network architecture that was introduced to computer vision through Vision Transformer (ViT) [8] from NLP. Transformer gained significant attention in natural language processing due to its self-attention mechanism, which excels at capturing long-range dependencies in text. ViT applied this mechanism to images by breaking them down into small patches and processing them as sequences, overcoming the limitations of traditional CNNs. Swin Transformer [9] further optimized this approach by introducing a locally aware self-attention mechanism, improving the efficiency of processing large-scale image data while preserving the ability to capture global context. Other examples of pure Transformer-based models include Segmenter [33], SegFormer [34], SwinUNet [35], and TransUNet [36].

However, to compensate for the Transformer's tendency to neglect local information, most existing methods usually employ hybrid designs that integrate CNNs. For instance, CMTFNet [37] proposed a fusion network combining CNN with multiscale transformer. This network leverages CNN's powerful local feature extraction capabilities alongside Transformer's multihead self-attention mechanism to effectively capture global context and model long-range dependencies. To address the gaps caused by conventional patches in Transformer, DBARCT [38] introduced parallel branches with irregular patches to enhance the model's ability to learn global context, improving its capture of spatial information.

In addition to improving the learning capabilities of Transformer, another consideration is how to effectively combine features from both Transformer and CNN. RoadCT [41] proposed a relational fusion block in decoder, which fuses local and global road features based on the importance between the CNN and Transformer blocks. This approach effectively combines the strengths of both architectures to enhance the

network's ability to capture and integrate relevant road features for improved segmentation performance.

C. Attention Mechanism in Road Extraction of HRSIs

As remote sensing segmentation tasks continue to evolve, models need to be capable of perceiving objects at different scales and excluding high-interference factors. Therefore, to satisfy this need, attention mechanisms are widely used in models with CNNs and transformers. For instance, the FEAM module proposed by Yu et al. [45] enhances the network's multiscale and global perception capabilities, aiding in road extraction tasks. The following two models apply attention mechanisms at the channel level to overcome background interference. Ai et al. [48] introduced a new channel attention mechanism module, MPGSE. By adding variance and coefficient of variation to the SE module, the model is enhanced against interference. The DCDE module in BMDCNet [49] is designed for both global and local multiscale context extraction. The SID-DA module is able to integrate global and local multiscale contextual information to make the extracted road structures more complete and continuous.

Additionally, channel attention and spatial attention can be combined to extract and enhance road features, as demonstrated in the CDAM module in RCFSNet [53]. To address the sparsity of road pixels, Dai et al. [52] introduced DAM module in RADANet. This module combines the sparse sampling capability of deformable convolutions with the spatial self-attention mechanism.

We proposed the OCGA module based on a graph structure to enhance features, particularly addressing the sparsity of road pixels. Attention mechanisms come in many forms, and our proposed MFS module functions similar to a gated attention mechanism. By passing features through the MFS module before segmentation, unnecessary features can be effectively suppressed, thereby improving segmentation accuracy.

III. METHOD

In this section, we first give an overview of the overall framework for road extraction and then describe the proposed MADSNet. Finally, the framework utilizes modules such as multiscale feature flexible extraction (MFFE) decoder, OCGA module, and multilevel feature selection (MFS) module to improve performance.

A. Overall Network Architecture

In the task of road extraction from remote sensing images, many high-performance network architectures are designed following the encoder-decoder structure. Our model is built on the classic encoder-decoder architecture to ensure efficient processing and transformation of input data. The overall architecture of the model is shown in Fig. 1. For the encoder, we utilize a pretrained ResNet34 model [54] as its core backbone structure, leveraging its powerful feature extraction capabilities to capture key information from the input data, thereby transforming it into more abstract and useful representations. Before decoding, the output features from the encoder are passed through the OCGA module for feature aggregation

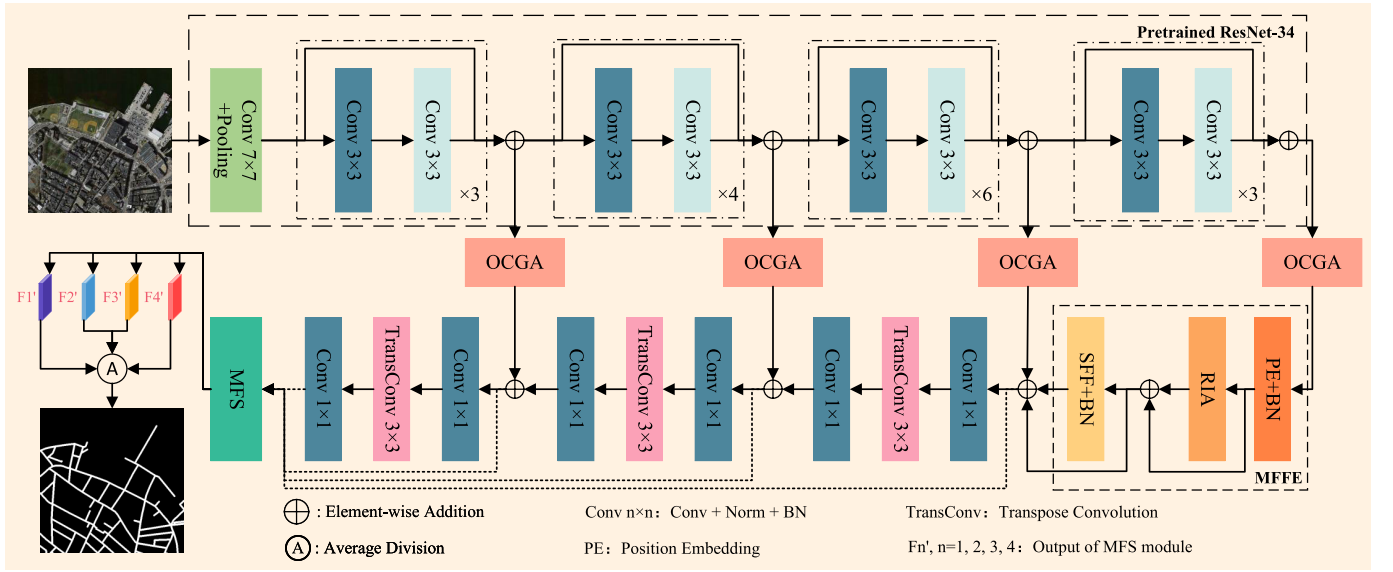


Fig. 1. Overall framework of the proposed MADSNNet. The encoder consists of the pretrained ResNet34. The structure of the MFFE decoder is enclosed in the dashed box at the bottom right. This includes two important modules: RIA module and SFF module. The features output from four-layer decoder are finally fed into the MFS Module.

and enhancement before being input into the decoder. OCGA module is a GAT improved by the KNNs algorithm, capable of capturing local information for each node and filtering out the most useful information based on importance weights. The features then flow into the decoder through skip connections. For the first layer of the decoder, we use the RIA, which is based on a window mechanism to decode the features. This allows the self-attention mechanism to better capture long-range dependencies, compensating for the limitation of OCGA, which can only capture local dependencies. The shifted window mechanism also helps reduce computational costs, especially when handling high-resolution images like remote sensing images. Next, the scope flexible fusion (SFF) module is connected to fuse the features. This module uses multiscale dilated convolutions for feature extraction. In the following three layers of the decoder, we use a CNN architecture for the second stage of decoding. The first layer of decoding already achieves a relatively high-quality result, and the subsequent decoding stages further refine the output. After all four decoding layers are completed, the features from these layers are fed into the MFS module. The MFS module performs feature selection and activation at different stages, and finally, the segmentation head outputs four results. These four results are averaged to produce the final binary road extraction map.

B. MFFE Decoder

In order to improve the model's ability to decode high-resolution images, we propose a MFFE decoder, primarily composed of two parts: RIA module and SFF module. The architecture of this module is shown in Fig. 2. RIA is a variant of the self-attention mechanism designed to obtain a more comprehensive representation of image features.

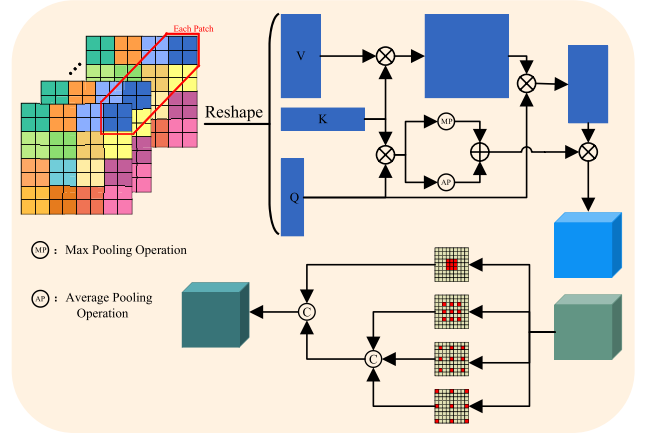


Fig. 2. Structures of two modules in MFFE decoder. The one above is RIA module and below is SFF module.

For the existing feature map $M_p \in \mathbb{R}^{H \times W \times C}$, attention computation is performed. For each patch, its features are transformed through deformable convolution to obtain $Q, K \in \mathbb{R}^{N \times C'}$, $V \in \mathbb{R}^{N \times C}$, where $N = H \times W$. The matrix multiplication between V and K results in $K^T \cdot V$, which contains the actual element information with correlated features. $K^T V$ is then multiplied by Q through matrix multiplication to obtain $Q \cdot K^T V$, which is denoted as M . We altered the original computation order from $(QK^T)V$ to $Q(K^T V)$. The change in the order of calculation results in two benefits: 1) one is that calculating K and V first reduces the sensitivity to position and makes each Q more effective in capturing the spatial dependence of similar pixels; and 2) another one is that it reduces the computational load without changing the result. The three matrices satisfy the associative property during the dot product, i.e., $(AB)C = A(BC)$, and calculating the two smaller matrices first helps lower the computation cost. If we

calculate QK^T first, the computation cost is $NC'N$, while if we calculate K^TV first, the computation cost is $C'NC$. Therefore, when C is smaller than N , the computation method we chose will result in a lower computational cost.

In order to simultaneously consider both the most salient features and general detailed features during feature processing, the matrix multiplication of Q and K produces $Q \cdot K^T$, which represents the attention scores of each element relative to the others. Max pooling is used to extract the attention scores of the most salient features from $Q \cdot K^T$, while average pooling captures a general representation of all relationships. Finally, element-wise addition of these can help the network focus on both the most relevant feature information and the overall contextual information ω . The calculation formula is

$$\omega = \alpha(Q \cdot K^T) \oplus \beta(Q \cdot K^T) \quad (1)$$

where $\alpha(\cdot)$ represents the max pooling operation, $\beta(\cdot)$ represents the average pooling operation, and $\oplus$ represents the element-wise addition operation.

The enhanced feature ωM is obtained by multiplying ω and M , and is then input into the SFF module. The SFF module is designed to enhance the model's ability to capture image details by effectively integrating feature maps of different scales, allowing the model to pay attention to both large-scale structural information and small-scale detail information when processing images. Specifically, we first perform multiscale feature extraction on the image using convolutional layers, and then feed these feature maps into three feature extraction branches with dilation rates of 1, 2, and 3, respectively. Finally, the features from these three branches are summed element-wise to complete the fusion of multiscale features. In addition, to enhance the model's generalization ability, we introduce multiple residual connections throughout the network architecture. Residual connections make it easier for the network to optimize during training, preventing issues of gradient vanishing or explosion. By adding the input feature map to the feature map processed through multiple layers, the model can retain more of the original information for improving overall performance.

C. OCGA Module

Typically, roads exist in the form of a road network, meaning that there are always other roads nearby. Therefore, using a graph structure to address road-related problems is quite appropriate. A graph consists of a set of nodes and a set of edges, where each node may represent a feature vector, and the edges may represent the weight attributes between nodes. GCN updates the state of each node by aggregating the features of the node and its neighboring nodes. This process usually involves a weighted summation of the features of neighboring nodes, with the weights typically determined by the strength of the connections between the nodes or the weights of the edges. As shown in Fig. 3, we propose a graph attention module based on the nearest-neighbor algorithm, which determines the neighboring nodes to aggregate by calculating the similarity between nodes as weights. GAT allow the network to learn the weights of the edges autonomously, and these edge weights can vary.

For a certain layer's encoder, the obtained features $R_g \in \mathbb{R}^{B \times C \times H \times W}$ are processed through a 3×3 convolution layer to obtain $R_h \in \mathbb{R}^{B \times C \times H' \times W'}$, where B represents the batchsize, C represents the number of channels, and H, W, H', W' represent the dimensions of the feature map. This process reduces the size of the feature map, thereby decreasing the computational load in the subsequent graph network. The feature map R_h is reshaped to obtain $R_i \in \mathbb{R}^{B \times N \times C}$, which is the multichannel pixel map of R_h , where $N = H' \times W'$. This transformation aligns with the computation rules for each pixel. According to the aggregation pattern of the graph network, each multichannel pixel map $R_i = \{P_i^{b_1}, P_i^{b_2}, \dots, P_i^{b_B} \in \mathbb{R}^{N \times C}\}$ must be calculated to obtain the updated attention map $AM_i^{b_j} \in \mathbb{R}^{N \times N}$, $j = 0, 1, 2, \dots, B$. For each batch's pixel map $P_i^{b_j}$, the Euclidean distance between each pixel and all other pixels (excluding itself) is computed to obtain $Q_i^{b_j} \in \mathbb{R}^{N \times N}$. The calculation formula is

$$d(P_i^{b_j}, P_{i'}^{b_j}) = \sqrt{(r_p - r_{p'})^2 + (g_p - g_{p'})^2 + (b_p - b_{p'})^2} \quad (2)$$

where the three-channel intensity values of pixel $P_i^{b_j}$ are $(R, G, B) = (r_p, g_p, b_p)$, and the three-channel intensity values of pixel $P_{i'}^{b_j}$ are $(R, G, B) = (r_{p'}, g_{p'}, b_{p'})$. Based on pixels $P_i^{b_j}$ and $Q_i^{b_j}$, the k most similar pixels to each pixel are selected as updating auxiliary objects. After the update, $S_i^{b_j} \in \mathbb{R}^{N \times k}$ is obtained, where each row of $S_i^{b_j}$ stores the coordinate positions of the k most similar pixels for each pixel. The calculation formula is

$$\text{Topk}(P_i^{b_j}, Q_i^{b_j}, k) = \{(x_1, y_1, z_1), (x_2, y_2, z_2), \dots, (x_k, y_k, z_k)\}. \quad (3)$$

Finally, the weights of these k pixels are set to 1, while the weights of all other pixels are set to 0. After obtaining $AM_i^{b_j}$, it is used in the neighbor node activation operation (NNAO) with R_i , allowing each pixel to activate its k most similar neighboring nodes based on $AM_i^{b_j}$. Let us assume we are calculating the attention score of node r during the NNAO, where $b \in \mathbb{N}_b$ represents the direct neighboring node of r . The attention score of r with respect to b is

$$a_{br} = \frac{\exp(\ell(h^T [W \vec{t}_b || W \vec{t}_r]))}{\sum_{b \in \mathbb{N}_b} \exp(\ell(h^T [W \vec{t}_b || W \vec{t}_r]))} \quad (4)$$

where $\vec{t}_b, \vec{t}_r \in R_i$ represents the pixel point vector in the feature map, $W \in \mathbb{R}^{C' \times C}$ denotes a shared learnable weight matrix, $h \in \mathbb{R}^{2C' \times 1}$ indicates a shared self-attention weight parameter, $(\cdot)^T$ represents transposition, $||$ is the concatenation operation, and $\ell(\cdot)$ is the LeakyReLU function.

At this point, based on $AM_i^{b_j}$, the weights of all neighbors except for the k most similar ones are set to an infinitesimal value, which can be set to -9^{15} . This approach allows them to be activated to 0 during the sigmoid activation operation. The graph convolution updates each node through this aggregation mechanism, and the updated nodes will carry the features of other nodes, thereby completing one graph convolution.

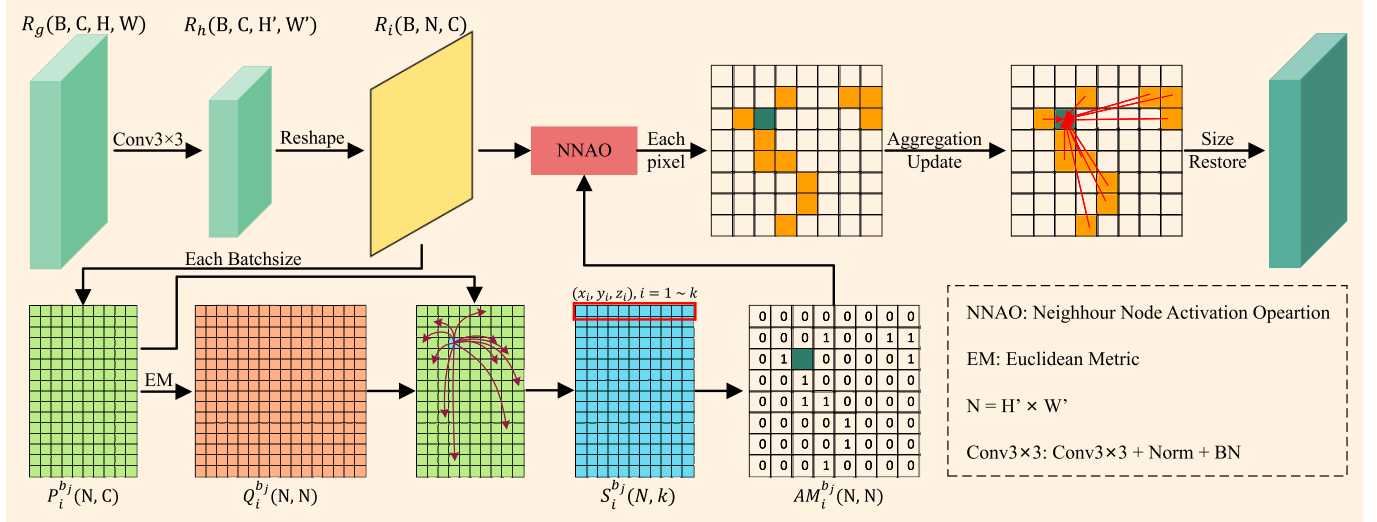


Fig. 3. Structure of the OCGA module. R_g is transformed in size to obtain R_i , after which the similarity between each pixel and all other pixels is computed. The top k most similar pixels are selected and neighbor features are aggregated using NNAO operation.

The features of the final output can be expressed as

$$t'_r = \delta \left(\sum_{b \in \mathbb{N}_b} a_{br} W t_b \right) \quad (5)$$

where $\delta(\cdot)$ is the ELU activation function.

After the graph convolution, it is necessary to restore the feature dimensions to their original size.

D. Multilevel Feature Selection Module

After passing through the four different layers of decoders, each stage has obtained feature maps of varying sizes. The low-level features contain richer global details, while the high-level features capture the overall structural information of the roads. To enhance the feature information at each stage, we propose the multilevel feature selection (MFS) module to focus on fusing the features from different stages and applying them back to each stage. The module structure is shown in Fig. 4. After feature fusion, filtering and activation operations are performed to eliminate noise and other interferences, allowing the model to concentrate solely on the information relevant to the current stage, thereby improving the understanding of contextual information.

The features D_i , $i = 1, 2, 3, 4$ from the four decoder layers are first reshaped and passed through a 1×1 convolution to obtain F_i , $i = 1, 2, 3, 4$, at which point all the features have the same size. The calculation formula is

$$F_i = \text{Conv}_{1 \times 1}(\varphi(D_i)) \quad (6)$$

where $\varphi(\cdot)$ is size transformation.

Concatenate F_i along the channel dimension to obtain F_m , and then use a 1×1 convolution to obtain the composite attention score map F_{mas} . This attention score map contains hidden features from all stages and different sizes, and through the subsequent activation step, the enhanced feature maps for each stage are output. The calculation formula is

$$F_{\text{mas}} = \text{Conv}_{1 \times 1}(\varepsilon(F_1, F_2, F_3, F_4)) \quad (7)$$

where $\varepsilon(\cdot)$ is concatenate operation.

The first step is to compute the attention weights. The shared attention map unit has two inputs, denoted as F_{mas} and F_i , and the output is f_i . The calculation formula is

$$f_i = \partial(\text{BN}(\text{Conv}_{1 \times 1}(\sigma(\text{BN}(\text{Conv}_{1 \times 1}(F_{\text{mas}})) \oplus \text{BN}(\text{Conv}_{1 \times 1}(F_i)))))) \quad (8)$$

where $\partial(\cdot)$ is sigmoid function, $\sigma(\cdot)$ is ReLU function.

The second step involves performing an element-wise multiplication of the attention weights f_i with F_{mas} , resulting in the activated feature J_i . The calculation formula is

$$J_i = f_i \otimes F_{\text{mas}}. \quad (9)$$

The third step is to merge the activated feature back into the original feature map to enhance it. Finally, a 1×1 convolution is applied to restore the original number of channels. The calculation formula is

$$F'_i = \text{Conv}_{1 \times 1}(\varepsilon(J_i, F_i)). \quad (10)$$

The parameters of the four small units that calculate the attention scores are shared, which reduces the complexity of the model and decreases the risk of overfitting. This weight-sharing mechanism also makes the model less sensitive to minor variations, thereby improving its robustness. Finally, these enhanced feature maps are passed through a segmentation head to obtain a binary image with a single channel. After averaging, the final segmentation result F_n is produced. The calculation formula is

$$F_n = \frac{\sum_{i=1}^4 (\xi(F'_i))}{4} \quad (11)$$

where $\xi(\cdot)$ is a division head with the number of output channels of 1.

IV. EXPERIMENTS

In this section, we first introduce three datasets: Google Earth [55], Massachusetts [56], and CHN6-CUG [57] road dataset, along with the experimental setup and evaluation criteria for various methods. Then, we compare the results

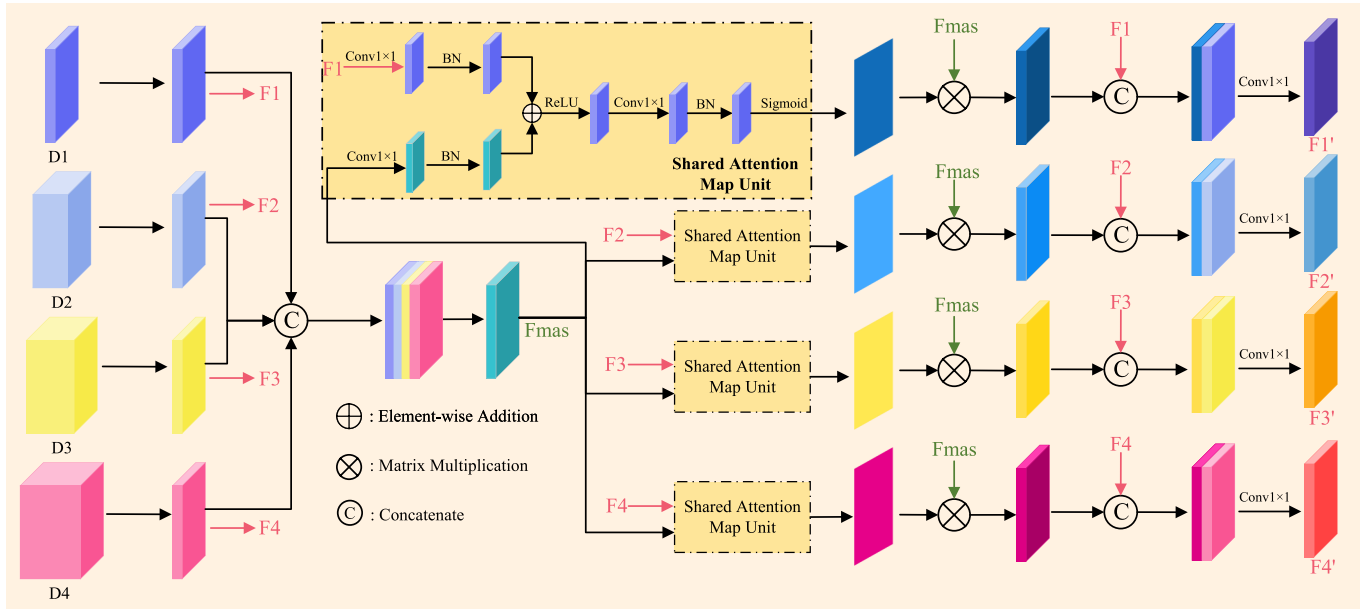


Fig. 4. Structure of the MFS module. The four output features from different decoders are concatenated to obtain F_{mas} , after which the features at each level are enhanced in parallel.

of our proposed method, MADSNet, with the current state-of-the-art methods. Finally, we conduct a detailed ablation study analysis of all the modules we proposed.

A. Dataset and Evaluation Metrics

1) *GoogleEarth Road Dataset*: The original spatial resolution of the Google Earth road dataset is 1.2 m/pixel. In this dataset, each image has at least 600×600 pixels. Roads in these images typically span about 12–15 pixels in width. Many of images contain intricate backgrounds and are often obscured by cars or trees, which makes road detection particularly difficult. The Google Earth dataset contains 224 images of different scales.

2) *Massachusetts Road Dataset*: The Massachusetts road dataset contains 1171 aerial images captured across the state. Each image has a size of 1500×1500 pixels, with a road resolution of 1 m per pixel. The dataset spans over 2600 km², covering diverse regions such as cities, suburbs, and villages, and includes roads, buildings, vegetation, vehicles, and more, with a coverage area of 2.25 km². The 1171 images consist of 1108 training images, 14 validation images, and 49 test images.

3) *CHN6-CUG Roads Dataset*: This dataset consists of images selected from six key cities in China: Beijing's Chaoyang area, Shanghai's Yangpu District, downtown Wuhan, Shenzhen's Nanshan area, Hong Kong's Shatin area, and Macao. Each image is 512×512 pixels in size, with a resolution of 50 cm/pixel. The dataset contains a total of 4511 images, divided into 3608 for training and 903 for testing.

In this pixel-level task, we use evaluation indicators such as intersection over union (IoU), precision rate, recall rate, and F1-Score to evaluate all the models in the experiment. These

metrics are defined by

$$\text{Precision} = \frac{TP}{(TP + FP)} \quad (12)$$

$$\text{Recall} = \frac{TP}{(TP + FN)} \quad (13)$$

$$\text{IoU} = \frac{TP}{(TP + FP + FN)} \quad (14)$$

$$\text{F1-Score} = \frac{2TP}{(2TP + FP + FN)} \quad (15)$$

where true positive (TP) represents the number of pixels correctly classified as road, while false positive (FP) refers to the number of nonroad pixels incorrectly extracted as road. True negative (TN) indicates the pixels accurately identified as background, and false negative (FN) represents the number of road pixels mistakenly classified as background. We also give a comparison of the computational and parametric quantities between the different methods. The parameter for the computational quantity is FLOPs and the parameter for the parametric quantity is Params.

B. Implementation Details

We used the ResNet (ResNet-34/ResNet-50) models as backbone networks and pretrained them on the ImageNet [58] dataset. The experiments are conducted using PyTorch 1.7.0 and Python 3.6.13. All experiments are performed on a workstation equipped with an Intel i7-12700 CPU, 16 GB RAM, NVIDIA GeForce GTX TITAN X GPU (with 12 GB of VRAM), and Ubuntu 20.04 operating system. We use the Adam optimizer, which combines the advantages of momentum and RMSprop (root-mean-square propagation) optimization methods. When updating parameters, Adam incorporates the first and second moment estimates of the

gradient, along with the learning rate, to calculate the parameter updates. For training on the GoogleEarth road dataset, the initial learning rate is set to 2×10^{-3} , with a weight decay of 5.0 applied at epochs [50, 80, 110]. For the Massachusetts and CHN6-CUG road datasets, the initial learning rate is set to 2×10^{-4} , with weight decay applied at epochs [70, 100]. Except for a few models, such as U-Net and CMTFNet, which can only have a batch size of 4 due to GPU memory limitations, the batch size for all other models is 8.

We use a weighted combination of dice loss and binary cross-entropy (BCE) loss as the loss function during the final model training. Dice loss is a metric used to measure the similarity between two sample sets, defined as

$$L_{\text{dice}} = \text{Dice}(y, \hat{y}) = 1 - \frac{2 \times |y \cap \hat{y}|}{|y| + |\hat{y}|} \quad (16)$$

where y represents the ground truth binary segmentation image, and $\hat{y}$ denotes the model's predicted segmentation result. A key feature of dice loss is that it penalizes both FPs and FN, making it particularly useful for handling imbalanced datasets, such as in road extraction tasks where the area of the background significantly exceeds that of the road area.

BCE loss is used for binary classification tasks and measures the difference between the model's predicted probability distribution and the target distribution. It is defined as follows:

$$L_{\text{bce}} = \text{BCE}(y, \hat{y}) = -[y \log(\hat{y}) + (1 - y) \log(1 - \hat{y})] \quad (17)$$

where y is the true label and $\hat{y}$ is the predicted result.

Therefore, the final total loss is defined as

$$L_{\text{total}} = \lambda_1 \cdot L_{\text{dice}} + \lambda_2 \cdot L_{\text{bce}} \quad (18)$$

where λ_1 and λ_2 are, respectively, 0.3 and 0.7.

C. Comparison With State-of-the-Art Methods

To verify the superiority of MADNet, we compare it with some popular methods on the GoogleEarth, Massachusetts, and CHN6-CUG Roads datasets. The selected methods mainly include convolution-based networks (e.g., U-Net [6], SegNet [10], LinkNet [11], D-LinkNet [59], DeepLabV3+ [12], and SGCN [46]), attention mechanism-based networks (e.g., MACUNet [60], RCFSNet [53], and SFFNet [50]), and transformer-based methods (e.g., CMTFNet [37] and CGGLNet [51]).

1) *Comparison With Other Methods on the GoogleEarth Road Dataset:* We compared the road extraction performance of MADNet on the GoogleEarth road dataset with several state-of-the-art methods, highlighting the best results in red. All methods used ResNet-34 as the backbone network. The GoogleEarth road dataset includes images with clearly defined road targets, making the overall segmentation relatively easy. The overall results for each method are shown in Table I. Our method achieved the highest accuracy, with an IoU of 88.51%, OA of 98.47%, precision of 94.26%, recall of 93.68%, and $F1$ of 93.87%. Compared to MACUNet, MADNet improved the IoU by 0.75%, indicating that there is still room for improvement in capturing interactions between the encoder and decoder using multiscale skip connections. MADNet

outperformed SegNet, which lacks diverse feature selection, by 3.85% in IoU. Furthermore, MADNet attained the highest IoU and $F1$ scores, thanks to its flexible feature decoder and multilevel feature selection module. Fig. 5 provides visual comparison examples. For this road dataset, we focused on the model's accuracy in handling details. As observed, nearly all methods successfully extracted the overall structure of the road network. In the first example of the first row, all methods except MADNet incorrectly identified the area within the red box as a road, showcasing a typical FP. The second and third examples demonstrate MADNet's effectiveness in handling densely packed road networks, where many models struggled to extract each road clearly. DeepLabv3+, RCFSNet, and SFFNet exhibited noticeable shadowing in the fourth example due to significant background interference. The remaining examples further illustrate MADNet's superior road continuity and robustness.

2) *Comparison With Other Methods on the Massachusetts Road Dataset:* Compared to the GoogleEarth road dataset, the Massachusetts road dataset presents greater challenges, with many roads obstructed by trees and buildings. Additionally, the surrounding background often resembles the roads, complicating detection. The overall results for each method are shown in Table I. It is important to note that while precision and recall should be as high as possible, their interdependence makes the $F1$ score more valuable for reference. Our MADNet achieved optimal performance with OA, IoU, and $F1$ scores of 98.00%, 66.14%, and 78.72%, respectively. Compared to LinkNet, MADNet improved IoU, and $F1$ by 0.97% and 0.79%, respectively. Although CMTFNet also utilized a multiscale M2SA module to extract global context and channel information, our MFFE decoder proved to be more effective, achieving increases of 0.49% and 0.38% in IoU and $F1$. Fig. 6 presents some visual output results. The areas within the green boxes indicate regions that all methods failed to detect, likely due to significant background interference or ambiguous labeling. The first example includes a small path heavily obstructed by trees, which our MADNet managed to extract effectively, while other methods either failed to detect it or could not delineate it completely. In the second example, all methods except MADNet missed certain roads. The third row features roads with colors and characteristics highly similar to the background, leading to significant issues of incompleteness and truncation for SegNet, LinkNet, DeepLabV3+, MACUNet, CMTFNet, and CGGLNet. The fourth row shows the segmentation of a path severely obstructed by trees and their shadows, which only MADNet and D-LinkNet managed to extract. In the final example, all methods except MADNet struggled to fully extract a branching path.

3) *Comparison With Other Methods on the CHN6-CUG Road Dataset:* Unlike standard datasets, the CHN6-CUG road dataset contains remote sensing road images from six different cities in China. This diversity results in a training set that does not meet the same distribution. Table I reports quantitative comparison results on the CHN6-CUG road dataset. The proposed MADNet achieved the highest scores of 95.68% for OA, 59.62% for IoU, 74.16% for precision, 73.67% for recall, and 71.82% for $F1$. Compared to U-Net and SegNet,

TABLE I

QUANTITATIVE COMPARISON OF ROAD EXTRACTION. THE BEST RESULTS ARE HIGHLIGHTED IN RED AND SUBOPTIMAL RESULTS IN BLUE. ALL RESULTS ARE EXPRESSED AS A PERCENTAGE (%)

Model	Backbone	FLOPs/G	Params/M	GoogleEarth					Massachusetts					CHN6-CUG				
				OA	IoU	Precision	Recall	F1	OA	IoU	Precision	Recall	F1	OA	IoU	Precision	Recall	F1
U-Net [6]	-	218.95	31.04	98.18	86.72	93.46	92.56	92.82	97.86	64.65	79.26	77.84	77.51	94.35	46.59	72.95	56.75	59.26
SegNet [10]	VGG16	160.52	29.44	97.88	84.66	93.00	90.78	91.63	97.85	63.87	79.90	75.89	76.93	94.69	48.79	72.61	60.12	61.48
LinkNet [11]	ResNet34	27.38	21.64	98.26	87.11	94.12	92.34	93.07	97.94	65.17	80.82	77.00	77.93	95.03	54.87	71.88	69.14	67.72
D-LinkNet [59]	ResNet34	30.63	11.25	98.30	87.36	93.21	93.51	93.21	97.91	65.36	80.01	77.80	78.15	95.25	55.39	72.89	69.57	68.37
DeepLabv3+ [12]	MobileNetv2	26.43	5.81	98.15	86.32	92.68	92.89	92.59	97.90	65.10	80.05	77.52	77.90	94.93	55.16	69.73	71.73	68.13
MACUNet [60]	-	33.59	5.15	98.39	87.76	93.66	93.47	93.42	97.95	65.63	80.53	77.75	78.30	94.72	51.48	70.59	65.36	64.33
SGCN [46] †	ResNet50	-	-	-	-	-	-	-	-	63.17	82.68	72.81	77.43	-	49.19	-	-	66.95
RCFSNet [53]	ResNet34	182.31	58.23	98.33	87.45	93.14	93.66	93.26	97.95	65.79	80.94	77.61	78.55	95.40	57.71	72.69	72.30	70.09
CMTFNet [37]	ResNet50	33.05	30.07	98.43	88.19	93.97	93.63	93.68	97.95	65.65	80.67	77.58	78.34	95.44	57.20	73.34	71.35	69.64
CGGLNet [51]	ResNet34	86.38	43.02	98.37	87.85	94.09	93.17	93.47	97.98	66.03	81.04	77.86	78.66	95.49	57.66	74.01	71.57	70.19
SFFNet [50]	ResNet34	31.92	24.13	98.38	87.79	94.16	93.04	93.45	97.97	65.82	80.35	78.15	78.42	95.51	58.80	72.95	73.42	71.22
Ours	ResNet34	34.34	28.99	98.47	88.51	94.26	93.68	93.87	98.00	66.14	80.98	78.00	78.72	95.68	59.62	74.16	73.67	71.82

* The results shown by the method marked with † are quoted from its related literature.

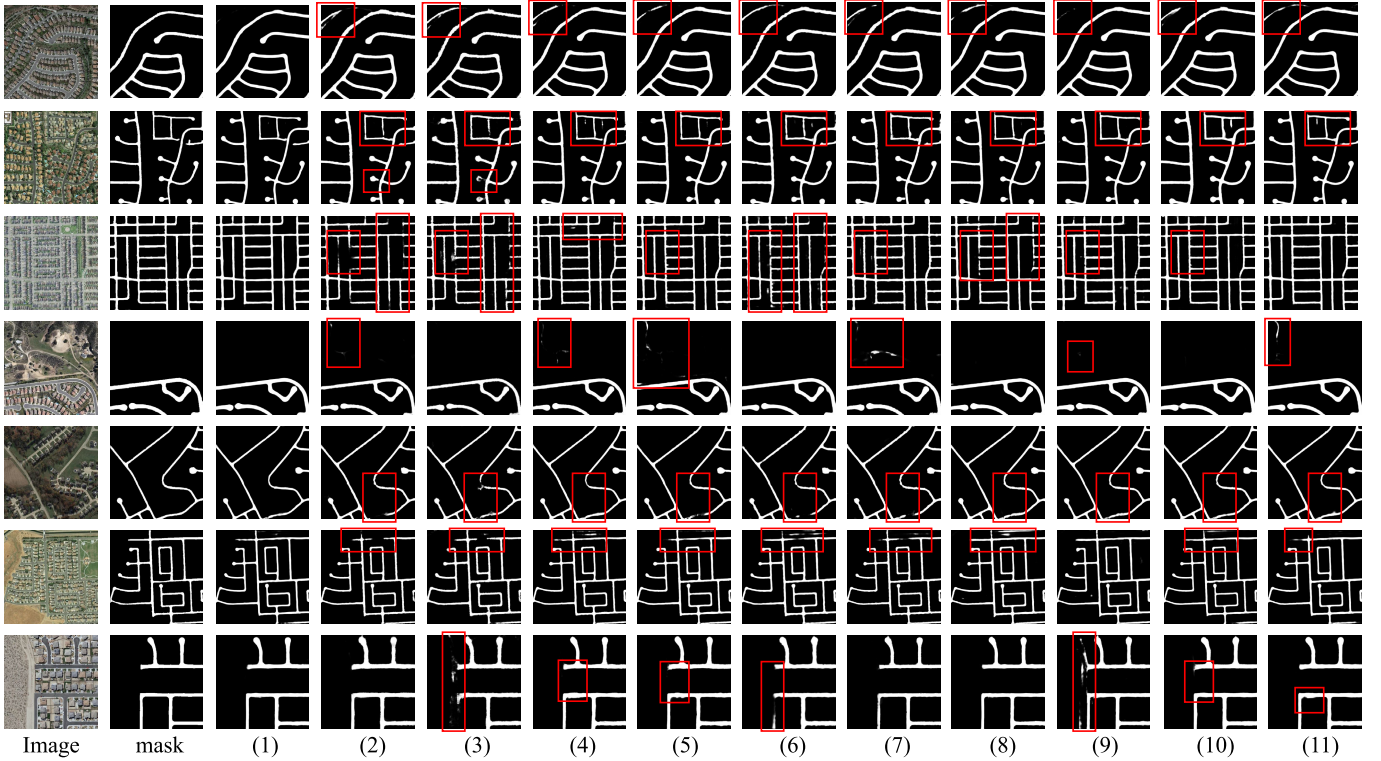


Fig. 5. Visualization comparison of different methods on the GoogleEarth road dataset. (1) Ours. (2) U-Net. (3) SegNet. (4) LinkNet. (5) D-LinkNet. (6) DeepLabv3+. (7) MACUNet. (8) RCFSNet. (9) CMTFNet. (10) CGGLNet. (11) SFFNet.

MADSNet improved IoU by 13.03% and 10.83%, respectively. Models based on attention and transformers significantly outperform those based solely on CNNs. Compared to RCFSNet, MADSNet improved IoU and $F1$ by 1.91% and 1.73%, indicating that dense skip connections based on dual attention have weaker generalization capabilities when handling data with different distributions. SGCN only aggregates encoder features at the highest level using depth-separable graph convolutions, while MADSNet improved its IoU by 2.94%. Compared to CGGLNet, MADSNet achieved increases of 1.96% in IoU, 0.15% in precision, 2.10% in recall, and 1.63% in $F1$. Fig. 7 displays visual results on the CHN6-CUG Road dataset. In this dataset, we primarily explore the robust capability of the proposed module in avoiding false detections. In the first and fourth examples, several methods misidentified background areas as roads. This issue becomes more pronounced in the

following examples. In the second example, all methods except MADSNet misclassified the red brick background within the yellow box as roads to varying degrees. In the third and seventh examples, nearly all methods mistakenly identified riverbanks as roads. In the fifth and sixth examples, due to the similarity in color between buildings and roads, all methods except MADSNet incorrectly classified the row-style residential buildings as road categories.

D. Ablation Studies

In this section, ablation experiments of the overall framework are implemented on the GoogleEarth, Massachusetts, and CHN6-CUG road datasets. The remaining more ablation experiments about the details of the component are performed on the GoogleEarth road dataset.

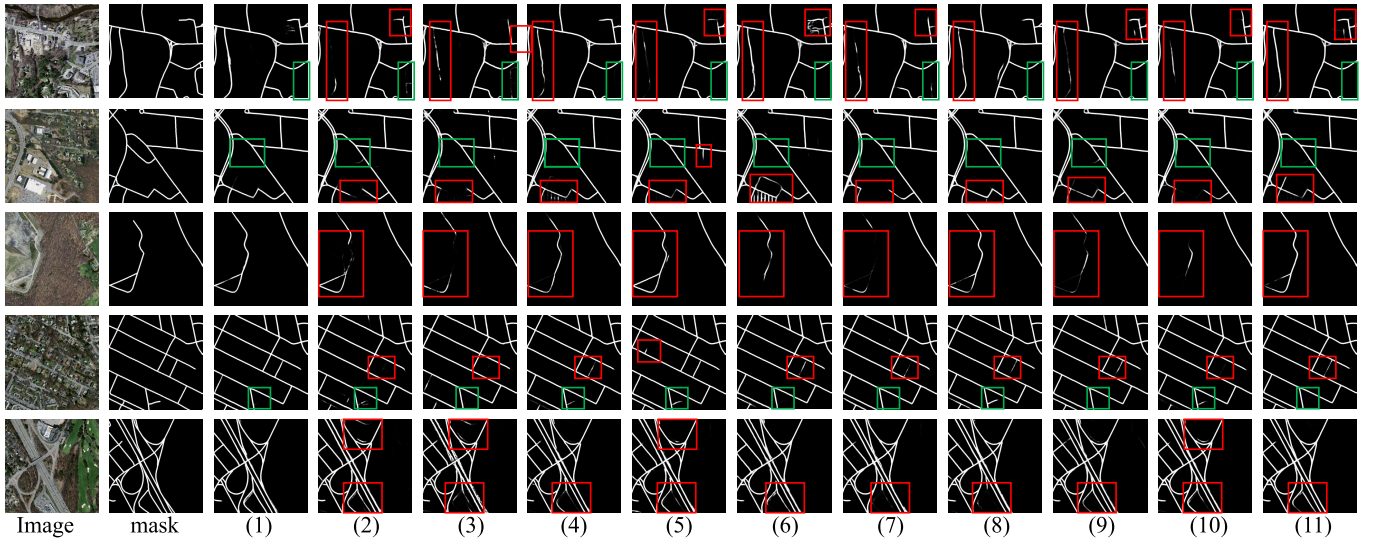


Fig. 6. Visualization comparison of different methods on the Massachusetts road dataset. (1) Ours. (2) U-Net. (3) SegNet. (4) LinkNet. (5) D-LinkNet. (6) DeepLabv3+. (7) MACUNet. (8) RCFSNet. (9) CMTFNet. (10) CGGLNet. (11) SFFNet.

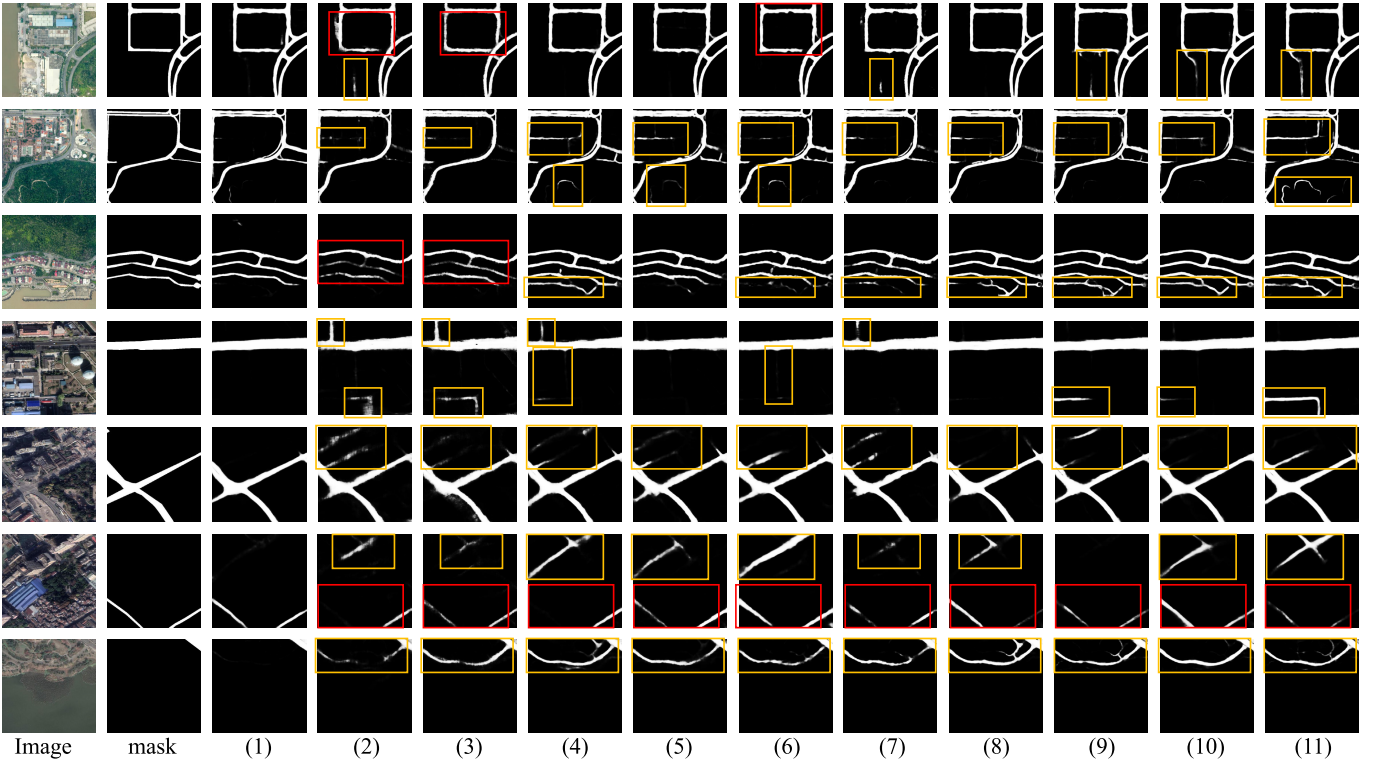


Fig. 7. Visualization comparison of different methods on the CHN6-CUG road dataset. (1) Ours. (2) U-Net. (3) SegNet. (4) LinkNet. (5) D-LinkNet. (6) DeepLabv3+. (7) MACUNet. (8) RCFSNet. (9) CMTFNet. (10) CGGLNet. (11) SFFNet.

1) *Effect of Each Module of MADSNet*: To thoroughly investigate the contribution of each module in MADSNet to the final performance, we conducted ablation experiments on each key module across three datasets. Baseline: The baseline is constructed by LinkNet with ResNet34 as the backbone network.

a) *Baseline + MFFE*: Four transposed convolution blocks are used as the decoder in the baseline. We replaced one of the highest-level transposed convolution blocks with

the MFFE decoder. As shown in Table II, by replacing the decoder, all performance metrics were improved. The attention mechanism significantly enhanced the model's ability to focus on important features, particularly when handling complex data, where its impact on accuracy was especially notable. On the CHN6-CUG road dataset, the network's IoU increased by 2.52%, precision by 3.04%, and $F1$ by 2.04%. This demonstrates the effectiveness of the MFFE decoder.

TABLE II

QUANTITATIVE COMPARISON OF THE EFFECTIVENESS OF EACH MODULE. THE BEST RESULTS ARE HIGHLIGHTED IN BOLD. ALL RESULTS ARE EXPRESSED AS A PERCENTAGE (%)

Dataset	MFEE	OCGA	MFS	OA	IoU	Precision	Recall	F1
GoogleEarth	✓			98.26	87.11	94.12	92.34	93.07
	✓			98.39	87.81	94.19	93.01	93.46
	✓	✓		98.42	88.08	94.07	93.41	93.61
	✓	✓	✓	98.47	88.51	94.26	93.68	93.87
Massachusetts	✓			97.94	65.17	80.82	77.00	77.93
	✓			97.95	65.50	80.44	77.59	78.22
	✓	✓		97.98	65.72	80.84	77.46	78.40
	✓	✓	✓	98.00	66.14	80.98	78.00	78.72
CHN6-CUG	✓			95.03	54.87	71.88	69.14	67.72
	✓			95.47	57.39	74.92	71.24	69.76
	✓	✓		95.53	58.45	74.10	72.79	70.88
	✓	✓	✓	95.68	59.62	74.16	73.67	71.82

b) *Baseline + MFEE + OCGA*: We used the decoder with MFEE. As shown in Table II, thanks to the graph attention mechanism of OCGA module, which enables the aggregation of neighboring nodes, the network achieved a 1.06% improvement in mIoU and a 1.12% increase in *F1* on the CHN6-CUG Road dataset, demonstrating the effectiveness of the OCGA module.

c) *Baseline + MFEE + OCGA + MFS*: The MFS module was introduced to build the complete MADSNNet network. As shown in Table II, MADSNNet achieved notable improvements on all three datasets, with IoU increases of 0.43%, 0.42%, and 1.17%, respectively. This is attributed to the MFS module's ability to enhance features at each level, making the extracted objects more prominent.

We also added a visualization experiment for the OCGA module. From the first row of Fig. 8(a), we can see that the road areas in the right image are much clearer, and some road branches that were blurred in the left image are now visible in the right image. This is because each pixel aggregates the features of the top k nearest neighbors with the most similar characteristics. As a result, there may be a few background nodes that are similar to road features, as seen in the upper right image of Fig. 8(a). The background contains sparse noise, which has a weak influence and does not affect the overall performance. The model is able to filter this out, as shown in the channel map in Fig. 8(b). All the brighter regions are marked as 0 after passing through the segmentation head. The second row of (a) shows that the surrounding areas of the road become more prominent after the weight aggregation in the OCGA module, which helps highlight the road region. This is also marked as 0 after passing through the segmentation head.

2) *Ablation of the MFEE Module*: The MFEE module consists of the RIA module and the SFF module. We conducted a series of ablation experiments focusing on two parameters in the RIA module related to attention: the number of attention heads and the window size. The number of heads was set to 2, 4, and 6, while the window sizes were set to 8 and 16. As shown in Table III, experiments with a window size of 16 generally performed better than those with a size of 8. The number of attention heads typically depends on the specific recognition or segmentation tasks; more categories to recognize require a larger number of heads, which also increases computational costs. In the road extraction task, the

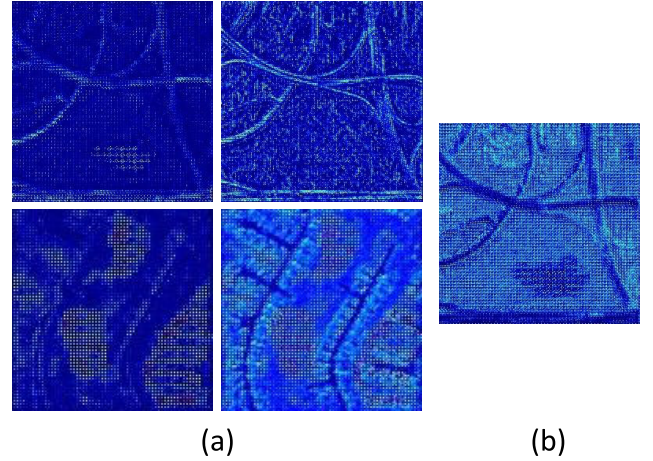


Fig. 8. Visualizations of output features with and without the use of OCGA module. The first row of (a) shows the channel visualization map focusing on the road areas. The second row of (a) shows the channel visualization map focusing on the areas around the road, such as houses and trees. (Left) Feature map without the OCGA module's output features, while (right) feature map with the element-wise added features after aggregation by the OCGA module. (b) Displays the channel visualization map related to the background.

TABLE III

QUANTITATIVE COMPARISON OF PARAMETER DETAILS OF MFEE MODULE. THE BEST RESULTS ARE HIGHLIGHTED IN BOLD. ALL RESULTS ARE EXPRESSED AS A PERCENTAGE (%)

Number of heads	Window size	OA	IoU	Precision	Recall	F1
2	8	98.37	87.76	93.73	93.42	93.43
4	8	98.41	88.05	94.19	93.28	93.60
6	8	98.41	87.98	93.26	94.12	93.55
2	16	98.38	87.81	93.69	93.51	93.46
4	16	98.47	88.51	94.26	93.68	93.87
6	16	98.45	88.14	93.32	94.24	93.64

categories to be segmented are relatively singular, leading to a decline in IoU and *F1* metrics when the number is set to 6. Compared to 2 heads, setting the number of heads to 4, respectively, improved IoU and *F1* by 0.7% and 0.41% when the window size is 16.

3) *Number of Selected Neighboring Nodes*: For the generation of the adjacency matrix in the OCGA module, we employed the k -nearest-neighbor algorithm, connecting the current node to its k most similar neighbors while treating the remaining nodes as disconnected. In the ablation experiments, we tested k values of 7, 9, 11, 13, and 15 to examine the impact of varying neighbor counts on the results. As shown in Table IV, increasing the number of neighbors improved IoU and *F1* to some extent. The best results were obtained at $k = 11$; however, as the number of neighbors continued to increase, the model's performance gradually declined. This decline is attributed to the inclusion of interference nodes that do not belong to the road category, which were also aggregated and learned by the graph convolution, leading to inaccurate predictions. Therefore, we set $k = 11$ for our experiments.

4) *Ablation of the MFS Module*: In the MFS module, there are four units responsible for computing the attention maps, all performing the same function and structured identically. Here, we first analyze and discuss whether to implement parameter sharing among these four units. Parameter sharing is

TABLE IV

QUANTITATIVE COMPARISON OF NUMBER OF SELECTED NODES IN OCGA MODULE. THE BEST RESULTS ARE HIGHLIGHTED IN BOLD. ALL RESULTS ARE EXPRESSED AS A PERCENTAGE (%)

Number of selected nodes	OA	IoU	Precision	Recall	F1
7	98.36	87.71	93.99	93.10	93.41
9	98.47	88.24	94.14	93.53	93.70
11	98.47	88.51	94.26	93.68	93.87
13	98.48	88.32	94.22	93.53	93.75
15	98.42	88.03	93.83	93.58	93.58

TABLE V

QUANTITATIVE COMPARISON OF PARAMETER SHARING IN MFS MODULE. THE BEST RESULTS ARE HIGHLIGHTED IN BOLD. ALL RESULTS ARE EXPRESSED AS A PERCENTAGE (%)

Parameter sharing	Fusion mode	OA	IoU	Precision	Recall	F1
Yes	Add	98.40	87.97	93.91	93.45	93.55
Yes	Concat	98.43	88.11	94.18	93.32	93.63
Yes	MFS	98.47	88.51	94.26	93.68	93.87
No	MFS	98.46	88.21	94.23	93.40	93.68

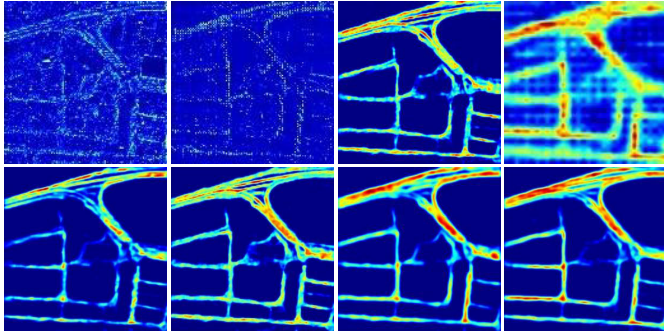


Fig. 9. Visualizations of output features of four different levels with and without the use of MFS module.

a common practice during model training, as it helps the model learn more generalized feature processing methods, maintaining stable performance across varying input sizes. Moreover, parameter sharing can reduce the number of parameters and lower the risk of overfitting. As shown in Table V, using parameter sharing improved IoU and *F1* by 0.3% and 0.19%, compared to not using it.

Next, we conducted ablation experiments using simple feature fusion methods for comparison. We directly performed element-wise addition of the decoder features from four different stages. As shown in Table V, the performance drops significantly compared to using the MFS module for fusion. When using feature concatenation for fusion, the performance degradation is relatively smaller. This is because feature concatenation better preserves the differences among features from different stages. We also provide the feature visualizations before and after applying the MFS module, as shown in Fig. 9. The first row shows the feature visualization results before applying the MFS module, while the second row shows the feature visualization results after applying the MFS module. From the images, it is clear that after passing through the MFS module, the road features have been significantly enhanced.

V. CONCLUSION

In this article, we propose MADSNet for HRSIs road extraction. MADSNet incorporates the MFFE module, composed of the relevance inquiry attention (RIA) module and the SFF module, which provides strong context information capture capabilities. RIA dynamically weights elements based on their relevance, improving the capture of long-range dependencies. The SFF module employs dilated convolutions to expand the receptive field, facilitating multiscale information fusion and retaining finer spatial details. Additionally, we introduce the SAF module, which activates and enhances features specific to each layer while suppressing noise and interference. These modules are highly portable and can be easily integrated into other models. Extensive ablation studies and comparative experiments across three public datasets demonstrate the effectiveness of MADSNet. In the future research, we will explore more universal modules and continue to work toward lightweight.

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